

Using Underwater Vision Profiler (UVP) data for Marine Conservation Areas monitoring

Report for the Department of Fisheries and Ocean (DFO) - St-John's, NL, Canada

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Introduction

The implementation of protection zones for biodiversity is one key area-based conservation tool, used worldwide to reach target 3 of the Kunming-Montreal biodiversity framework^{1,2}. For the ocean, Canada follows the international objective of protecting 30% of marine ecosystems by 2030, through the creation of Marine Protected Areas (MPA) and other effective area-based conservation measures (OECMs)³. Qualifications of conservation areas differ (e.g. Protected areas, National or Provincial Parks, Sanctuaries, Fishery Closures), but all these zones are spatially defined, protected year-round, and actively managed by an authority⁴. In this work, we will refer to them as “conservation areas” for simplicity.

Plankton - thousands of species from all kingdoms drifting in the pelagic realm - have an immeasurable value for ecosystems and humanity⁵. They form the foundation of aquatic food webs, play key roles in climate regulation, influence water quality and provide artistic inspiration⁶. In monitoring programs of conservation areas, plankton is often characterized *a posteriori* with a view to an ecosystem-based approach of conservation⁷⁻⁹. However, they are generally not considered during the establishment of conservation areas, mostly directed towards charismatic, and primarily vertebrates, species^{10,11}. In the sub-Arctic and Arctic regions of Canada, marine conservation areas seek to protect benthic fragile diversity (coral, sponges) and species of high nutritional and cultural value threatened by fishing activities^{3,12} and climate change¹³. However, while anthropogenic activities reach all depths of the ocean,

some deep and pelagic zones are underrepresented in marine protection¹⁴, leaving the largest volume of ocean where plankton thrive unprotected. A need for a 3D thinking of ocean sustainability is advocated¹⁴, taking into account plankton biomass, diversity and distribution^{15,16}.

One of the main challenges lies in the very high spatial and temporal variability of plankton dynamics^{6,16}, which makes it difficult to derive reliable *in situ* metrics from spot net sampling over large areas. Recent imaging devices enable improved characterization of this bloomy and patchy diversity¹⁷ by increasing sampling scales, owing to their ease of deployment, high acquisition frequency, and automated data analysis^{18–20}. Among them, the Underwater Vision Profiler (UVP) quantifies the abundance of particles (>100 µm) along depth profiles, while taking images of larger objects (>700 µm)²¹, including zooplankton and detritic particles. As plankton produces particles in the form of mucus wastes, fecal pellets and aggregates, the characteristics of particles (size, aspect..) will vary depending on the composition of the surrounding community²². In other words, the type of particles seen from images could be an indication of the structure of the pelagic biodiversity.

Following the general need to better monitor pelagic structure in marine conservation areas, this report aims at guiding the computation of UVP metrics for the conservation programs of Newfoundland and Labrador (Department of Fisheries and Ocean, St-John's, NL, Canada). Indeed, some data produced by the UVP require significant post-processing to derive ecological information, especially for images classification that need human validation assisted by machine learning tools²³. Hereafter are presented metrics that could be routinely measured during marine conservation programs, requiring minimal processing while aiming to provide meaningful ecological information on ecosystems.

The report will:

- (i) give information and resources about UVP functioning;
- (ii) present different easy-to-use UVP metrics and associated literature;
- (iii) provide codes for calculating them;
- (iv) test the codes on one dataset from Newfoundland and Labrador marine conservations areas program and;
- (v) provide recommendations and suggestions for conducting more advanced plankton monitoring.

UVP functioning

The Underwater Vision Profiler (UVP6 is the last version) is a tool designed by researchers and engineers of the Laboratoire de Villefranche-sur-mer to quantify large particles and zooplankton (100 µm - 2000 µm) by imaging and counting them in a fixed volume of water. It is classically deployed attached to a rosette, taking up to 20 images per seconds, at the intersection between the camera field of view and a red-light beam. All particles (> 100 µm) are counted according to their size range, while images in black and white are kept for the largest one (> 620 µm). After its deployment, the data is downloaded, processed and associated with metadata with the help of *UVPApp2* software. Only the images associated with the descent of the rosette are kept; it corresponds to an in-depth UVP cast, also referred to as a sample or a profile. Villefranche's team hosts a [website](#) with all resources of the UVP: [manual and videos](#) describing UVP deployment and the [UVP-App software installer](#).

Data is then transferred in two web applications: *Ecopart* and *Ecotaxa* (procedures also detailed in a [manual](#)).

- *Ecopart* is dedicated to the storage and management of particle data from the entire size range, providing counts and biovolumes of particles according to size ranges.
- *Ecotaxa* allows the storage and taxonomical classification of images (particles and zooplankton > 620 µm). It provides tools for the classification of images assisted by a random forest algorithm: categories can be created for non-living particles (ex: detritus) and for living plankton, now following WORMS classification and with a recent guide published, EcoTaxoGuide²⁴. Then, the algorithm suggests a classification of images based on a learning set already sorted: the prediction step. However, the main limitation is that in order to obtain a high-quality classification that can be used for biodiversity counting, a manual validation step for the categories is necessary.

In this report, we present only metrics that can be computed without any classification, or at the prediction step of the classification, but provide meaningful ecological information.

Download raw data from Ecotaxa and Ecopart

-> see the "Guide - Use Ecotaxa and Ecopart to export data" associated to this report.

From raw data to UVP metrics with a R code

Statement: The metrics computed here will undoubtedly provide valuable insights into pelagic ecosystems. However, they have not been produced by a scientific consortium and should therefore be used with caution. Further studies would be beneficial to validate their relevance and use.

Horizontal resolution: all metrics and scores are computed per cast, *i.e.* per UVP deployment

Vertical resolution:

- Particle-related metrics (Table I: m1, m2, m3, m4) were calculated within the first 200 meters of the water column, because (i) this zone is usually the densest in particles and (ii) it avoids a bias linked to station depth. Indeed, as stations at greater depths have more depth intervals with few particles, their average particle quantities will be lower. These metrics are calculated for particles ranging from 128 to 8200 µm, since particles outside this range are inaccurately quantified by the UVP.

- Plankton-related metrics (Table I: m5, m6) were averaged over the first 50 meters and last 50 metres of the water column, to follow the bi-modal distribution of the organisms.

R code: the associated R code (*Code - UVP_metrics_from_raw_data.R*) computes six metrics chosen from literature review (Table I).

Metrics explanation

The **average particle density per cast (m1)** is the average number of particles per liter observed by the UVP across the 0-200m depth profile. This metric reflects the average number of elements, living or non-living (marine snow or plankton), observed in the water column and forming the basis of marine food webs^{25,26}.

The **average biovolume of particles per cast (m2)** is the average volume of particles in mm³ per liter observed by the UVP across the 0-200m depth profile. This metric is similar to the average particle density, but is associated with a volume, giving the proportion of the water column filled with particles and a clearer indication of the amount of carbon available at the base of the food web^{27,28}. However, it is very sensitive to the presence of large particles, which are under-sampled by the UVP. The values associated with each profile must therefore be interpreted with caution.

The **slope of the relationship between particle abundance and size (m3)** for each cast is an indicator of ecosystem functionality and flows^{29,30}. For each particle size range, the concentration is compared to the size of the middle of the range. The relationship between the log of these two metrics is generally decreasing, and we can therefore derive the slope of this linear relationship, which takes a negative value located globally between -5 and -2. The closer the value is to 0, the less steep the slope, and the greater the concentration of large particles relative to the concentration of small particles. This is indicative of significant transfer within food webs and, therefore, a productive ecosystem. The maximum particle size used for this metric is 4100 µm instead of 8200 µm, as the linearity of the relationship is stronger in that range^{27,29}.

The **Shannon morphological diversity index (“morpho-Shannon”, m4)** is a diversity index of particle morphologies (concepts are exhaustively described in Trudnowska et al. 2021²² and Beck et al. 2023³¹). More precisely it quantifies the equitability of particle abundances within 5 morphological clusters (dark, elongated, fluffy, flakes and agglomerated particles). As particles are produced by diverse phyto- and zoo-planktonic organisms, it can be considered as a proxy of the pelagic diversity. This index is constructed by following the steps outlined in Trudnowska et al. (2021), adapted so that morphological variations of new images are described in relation to a fixed reference:

1. Selection of 24 morphological descriptors routinely computed from images during each UVP deployment
2. Mathematical transformations to make their distribution normal and remove outliers
3. Use of a reference morphological space (a Principal Component Analysis) created with typical images of the area of interest (past project in Baffin Bay and the Labrador Sea) to project the new UVP images taken in the conservation area as “additional individuals”. Their morphological variations are summarized in four dimensions of a PCA similar to that presented in Trudnowska et al. (2021).
4. Each new image is assigned to one of these clusters by calculating the minimal Euclidean distance between their coordinates in the morphological space and ones of the reference cluster barycenters.
5. At each station, the concentration of images per cluster is used to compute a morpho-Shannon index (clusters are used as species). As the number of species is fixed (5), the final morpho-Shannon index reflects the equitability of the distribution of particles within

clusters. The index varies between 1 and 3, with a high index indicating that all types of particles are present in similar proportions and a low index revealing that one or few types of particle predominates.

Copepod density (m5) is the number of images predicted as copepods in the first 50 metres and last 50 metres of the water column, ponderated by the sampled volume. Copepods dominate the zooplankton biomass of Arctic and sub-Arctic ecosystems and are hubs of matter and energy in food webs³². Their biomass is strongly correlated with that of total mesozooplankton in the North Atlantic and Arctic^{33,34}. Their depth distribution is mainly bimodal, with a mode close to the surface and another one close to the bottom³⁵. Therefore, this metric target averaged copepod densities at these two depth ranges. To obtain their abundances with minimal processing, we rely on the images predicted in the "Copepoda" categories (all order, families and genus levels) from the machine learning algorithm of Ecotaxa by using a learning set of reference for the (sub)Arctic ecosystem built by L. Vilgrain and its team at Université Laval. Copepods are the zooplankton taxa that is the best sampled by the UVP in the region¹⁸ and to ensure the reliability of the prediction, we tested its performance for copepods classification on another entirely validated dataset of the same region. We chose to trust Ecotaxa predictions because:

- there was a very significant linear relationship between the number of copepods manually validated and the predicted number of copepods (p-value < 0.001, R-square = 0.98);
- there was a slight over-estimation: the number of predicted copepods was 1.07053 times the number of validated copepods;
- on a dataset of 37451 objects including 6829 were copepods: 6432 were correctly identified (83%), 397 were missed (False negative), 904 were classified as copepods even if they were not (False positive), and 29718 objects were correctly placed to other categories (True negative): the overall accuracy of the model for copepods was 96%.

Large copepod density (m6) is the same metric but associated with copepods of a length >2mm, thus corresponding to *Calanus* copepods in (sub)Arctic ecosystems³⁶. Looking at this variable is really important for food webs as these organisms accumulate large lipid reserves^{37,38}, of main importance for their numerous predators such as the bowhead whales³⁹, cods⁴⁰ and birds⁴¹.

Table I. List of metrics (1 to 6) along with their units, the web application from which raw data need to be downloaded and references about their use and calculation.

	UVP metric	Unit	Web application	References
m1	average density of particles (128-4100 µm) per cast	#.L ⁻¹	Ecopart	Forest <i>et al.</i> , 2012 ²⁵

m2	average biovolume of particles per cast (128-4100 μm)	$\text{mm}^3 \cdot \text{L}^{-1}$	Ecopart	Clements <i>et al.</i> , 2022 ²⁷
m3	slope of the relationship between particles abundances and size (128-4100 μm)	NA	Ecopart	Kiko <i>et al.</i> , 2022 ²⁹
m4	shannon morphological diversity index from images (>620 μm)	NA	Ecotaxa No sorting needed	Trudnowska <i>et al.</i> 2021 ²² Beck <i>et al.</i> 2025 ³¹
m5	copepods density (close to surface and sea bottom) (>620 μm)	$\text{ind} \cdot \text{L}^{-1}$	Ecotaxa NB: need a constant learning set for prediction	Aarflot <i>et al.</i> 2018 ³³ Carstensen <i>et al.</i> 2019 ³⁴
m6	large copepods (>2mm) (close to surface and sea bottom) density	$\text{ind} \cdot \text{L}^{-1}$	Ecotaxa NB: need a constant learning set for prediction	Enstipp <i>et al.</i> 2018 ⁴¹ Bouchard <i>et al.</i> 2020 ⁴⁰ Fortune <i>et al.</i> 2020 ³⁹

Metric recommendations and final scores

Table II. Relevance of metrics (1 to 6) for ecosystem monitoring with estimation of their Processing difficulty and time (PDT), and Ecological significance (ES).

	UVP metric	PDT* (/5)	ES** (/5)	ES/ PDT	Relevance for ecosystem monitoring	Ref
m1	average density of particles (128-4100 μm) per cast	1	1	1	easy to compute but abundance is less relevant than biovolume for energy transfers in food webs	Forest <i>et al.</i> , 2012 ²⁵
m2	average biovolume of particles per cast	1	2	2	give information on the volume occupied by pelagic material and part of the food	Clements <i>et al.</i> , 2022 ²⁷

	(128-4100 μm)				web: kept in final abundance score	
m3	slope of the relationship between particles abundance and size (128-4100 μm)	3	5	1.7	give information on ecosystems productivity through size distribution: kept in final abundance score	Kiko <i>et al.</i> , 2022 ²⁹
m4	shannon morphological diversity index from images (>620 μm)	4	4	1	provides information on the equitability of particle morphologies: kept as final diversity score	Trudnowsk a <i>et al.</i> 2021 ²² Beck <i>et al.</i> 2025 ³¹
m5	copepods density (close to surface and sea bottom) (>620 μm)	3	5	1.7	provides information on the abundance of key zooplankton taxa: kept in final abundance score	Aarflot <i>et al.</i> 2018 ³³ Carstensen <i>et al.</i> 2019 ³⁴
m6	large copepods (>2mm) (close to surface and sea bottom) density	3	4	1.3	provide information on the abundance of a subset of large zooplankton: ecologically significant but redundant with m5	Enstipp <i>et al.</i> 2018 ⁴¹ Bouchard <i>et al.</i> 2020 ⁴⁰ Fortune <i>et al.</i> 2020 ³⁹

1. Pelagic Score 1 - Particles and plankton abundance score (m2, m3, m5)

The Pelagic Score 1 is calculated for each cast from the metrics of biovolume (m1), slope (m2) and copepod density (m5) to give an overall value for the quantity of plankton and particles observed. To calculate the score, each of the three indices is scaled between 0 and 3. The minimum and maximum values corresponding to 0 and 3 are determined from values typically found in the literature. For biovolume, the limits are set between 0 and 10 mm³.L⁻¹ (Clements *et al.*, 2022)²⁶; for slope, the bounds are set between -5 and -3 (Kiko *et al.*, 2022)²⁹; for copepod density, the bounds are between 0 and 5 ind/L, based on densities observed from a past UVP6 project in Baffin Bay and the Labrador Sea. Finally, the score is an average of the 3 scaled indices. This score is bounded between 0 and 3: a high index indicates a high quantity of particulate matter produced by, or available for, food webs; a low index indicates relatively lower quantity of particulate matter produced by, or available for, food webs.

N.B.: The scale of this score theoretically allows for the comparison of casts between different MCAs, however this comparison is difficult as it is relative to the timing and environmental conditions associated with sampling, which have a considerable impact on plankton distribution. It is therefore preferable to use it to compare casts related to a mission sampling one MCA.

2. Pelagic Score 2 = Particles and plankton diversity score (m4)

The **Pelagic Score 2** is the **morpho-Shanon diversity index (m4)** considered as a proxy of pelagic diversity. It quantifies the equitability of particle abundances within 5 morphological clusters (dark, elongated, fluffy, flakes and agglomerated particles). **This score is bounded between 0 and 3: a high index indicates that all types of particles are present in similar proportions and a low index reveals that one or few types of particle predominates.**

Preliminary results for a currently existing dataset

The Hawke Channel Closure is an 8,837 km² marine refuge located off the South-East coast of Labrador. The area is divided into 3 distinct bathymetric regions (Figure 1): the South-West region corresponding to the West margin of the trough and close to the Labrador coast, the North-East region which is a bank limiting the trough at the East, and the middle region that corresponds to the passage of the trough.

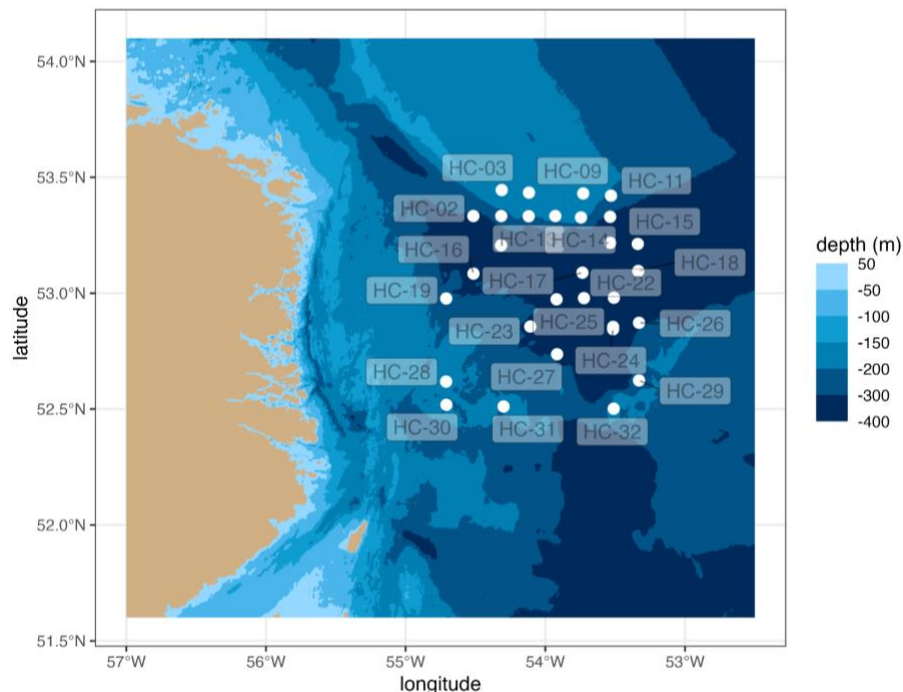


Figure 1. Bathymetry of the sampling area (Hawke Channel Closure).

The particles and plankton assemblages revealed by the UVP metrics during the sampling mission of September 2024 show a distribution related to the 3 regions described above:

- The **South-West region** is characterized by high particle densities (m1), rather low particle biovolumes and slopes (size-density relationship) (m2, m3), and low copepod and large copepod densities (m5, m6).

- The *middle region* is characterized by low particle density (m1), medium to high particle biovolumes and slopes (m2, m3), and medium copepod and large copepod densities (m5, m6).
- The *North-East region* is characterized by rather low particle densities (m1), medium to high particle biovolumes (m2), and high slopes, copepod and large copepod densities (m3, m5, m6).

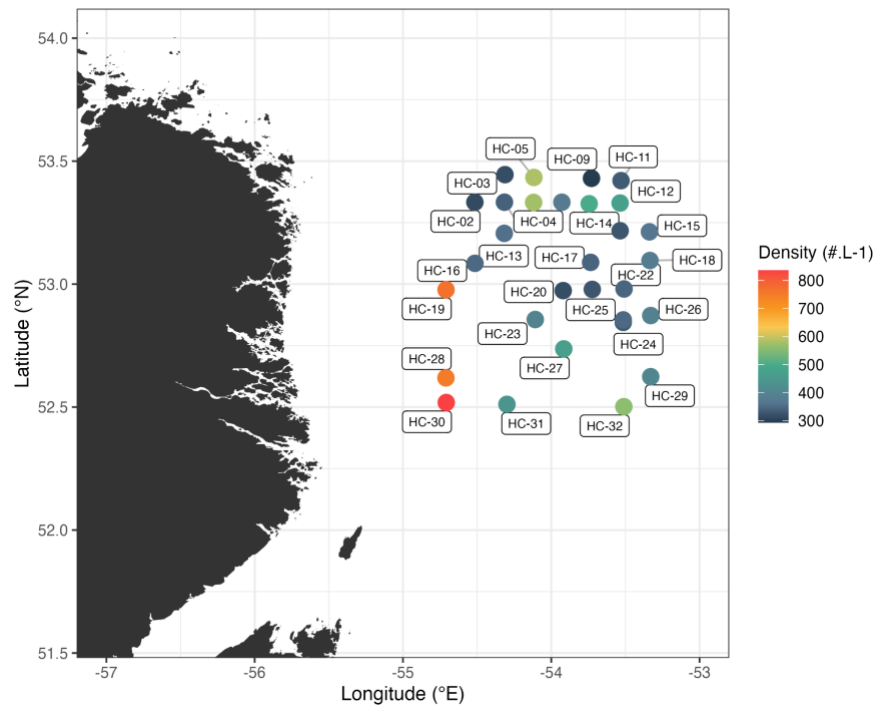


Figure 2. Average particle density (m1, #/L) in the first 200m of the sampled sites.

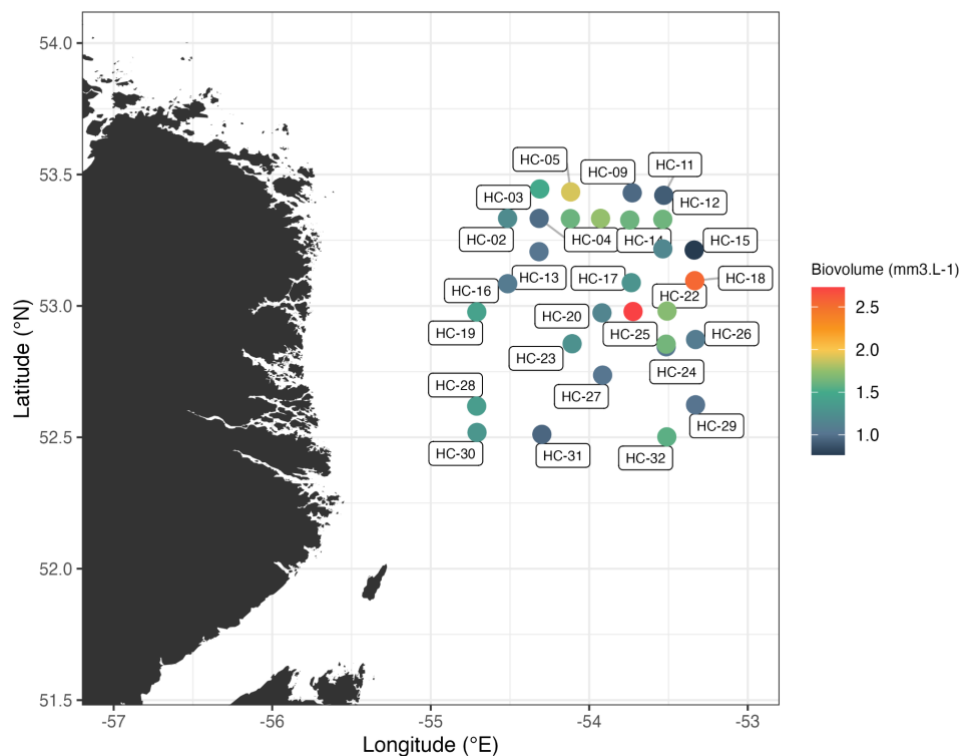


Figure 3. Average particle biovolume (m2, mm³/L) in the first 200m of the sampled sites.

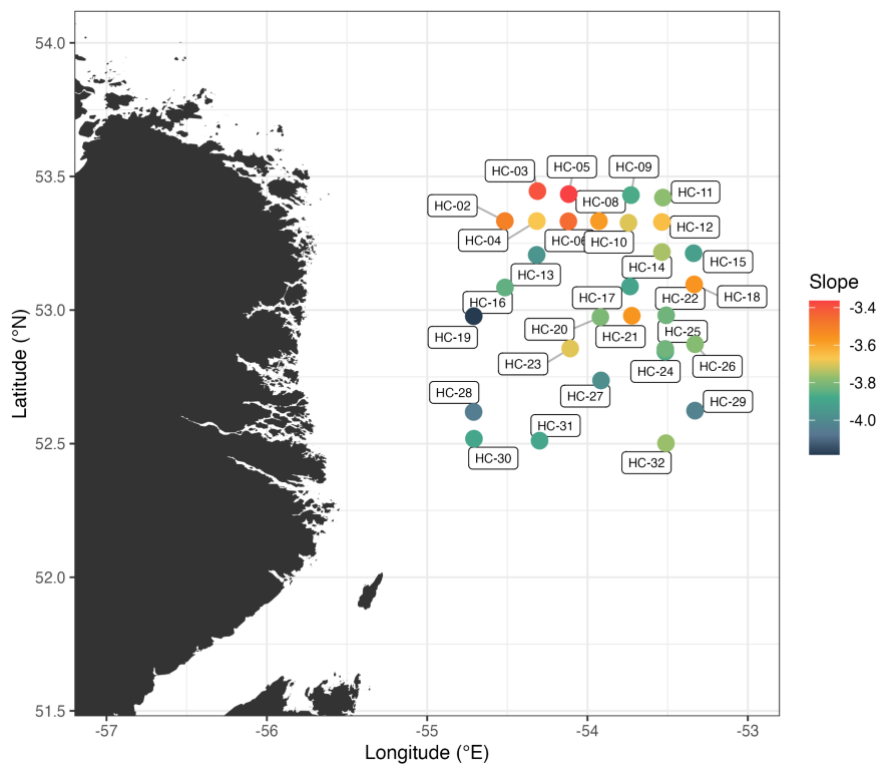


Figure 4. Slope of the relation between particles concentration and size (m^3), in the first 200m of the sampled sites.

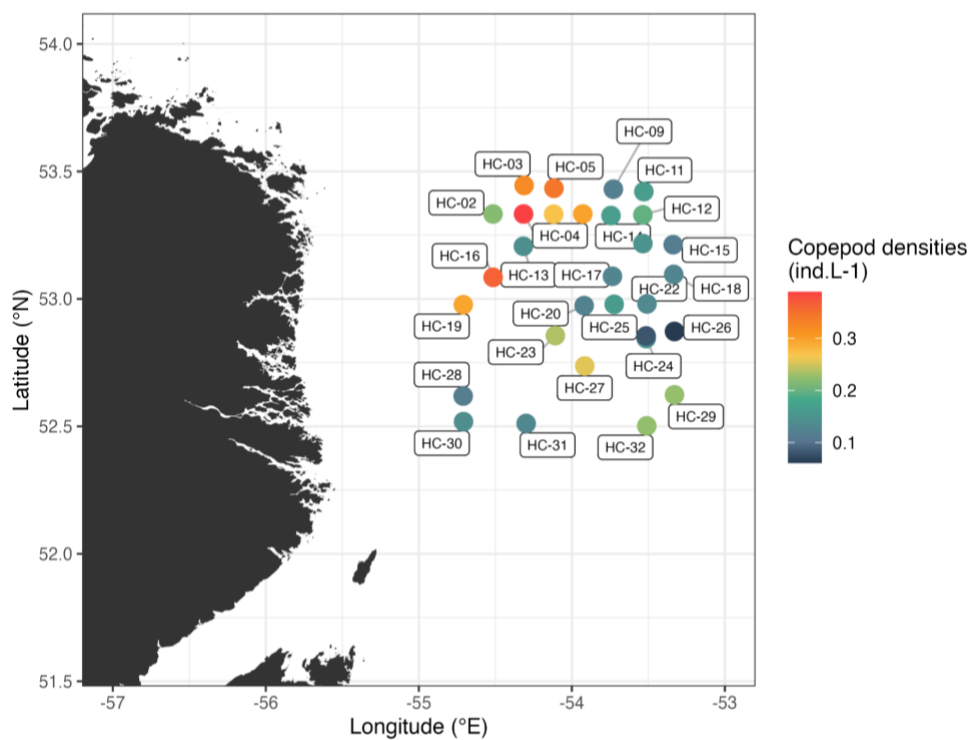


Figure 5. Averaged copepod densities (m^5 , #/L) in the first and last 50 meters of the different sampled sites.

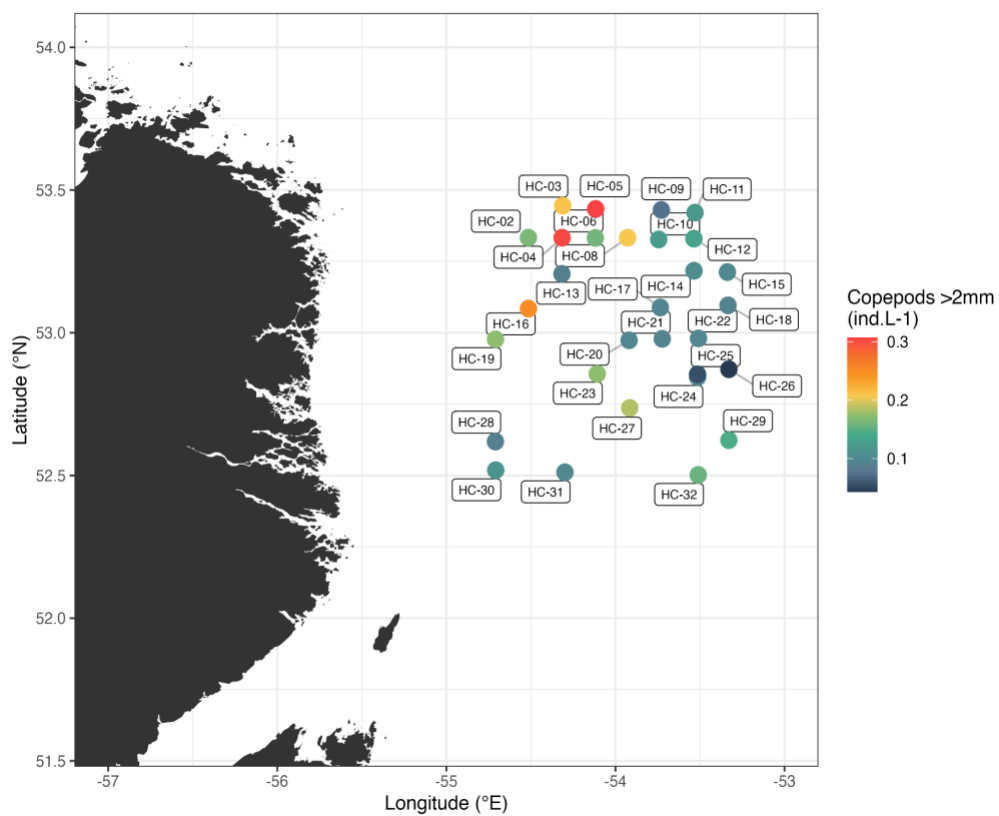


Figure 6. Averaged large copepod (>2mm, likely *Calanus* species) densities (m6, #/L) in the first and last 50 meters of the different sampled sites.

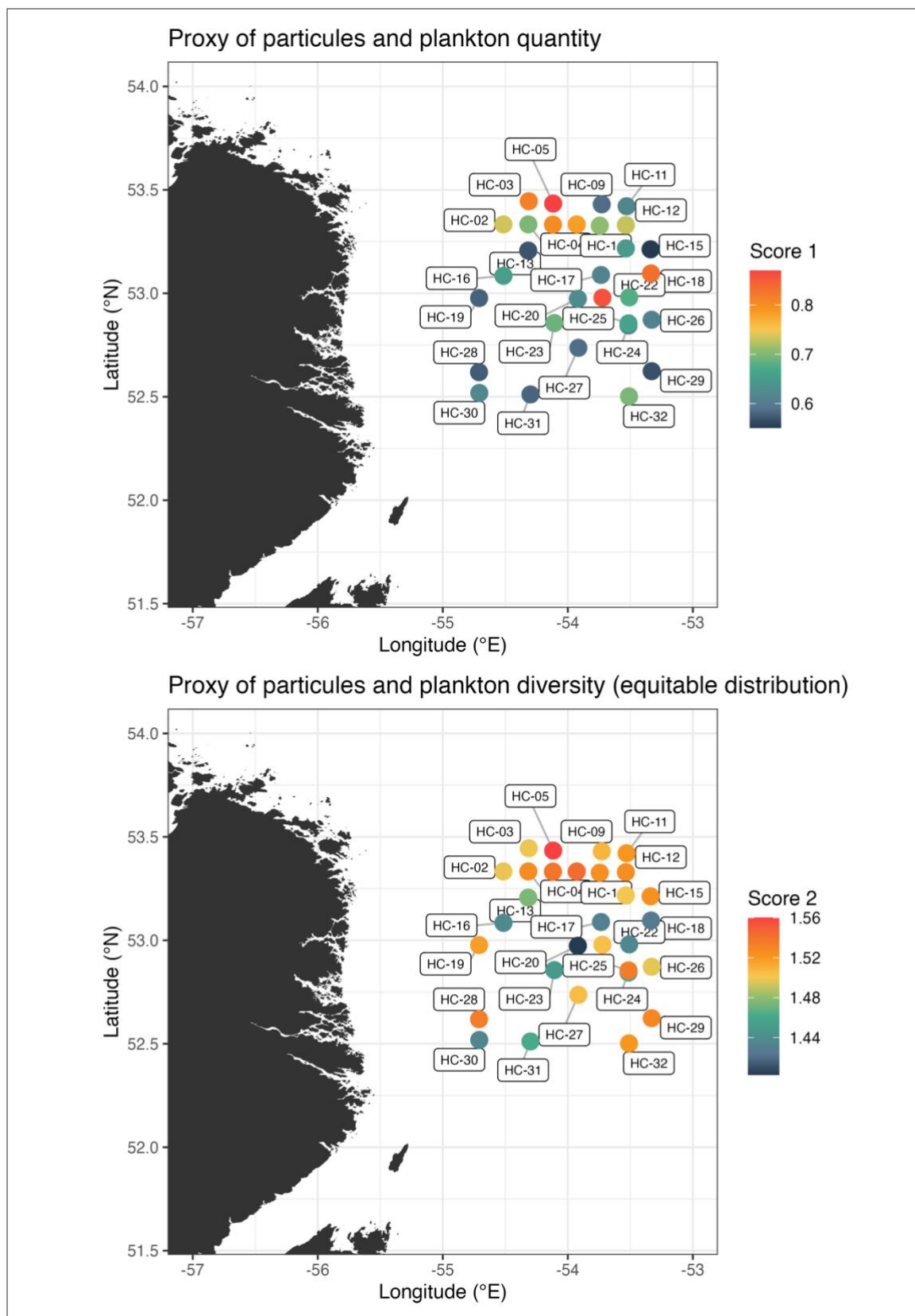


Figure 7. Score 1 is proxy of particles and plankton quantity (using m_2 , m_3 and m_5) and Score 2 (equal to m_4) is a proxy of particles and plankton diversity (more precisely: the equitability of particle abundances within 5 morphological clusters).

Preliminary conclusions

The *South-West* region shows higher densities of small particles, but lower meso-zooplankton biomass; while *middle* and particularly *North-East* regions tend to show more productive planktonic ecosystems and high copepod abundances, reflected in Score 1.

It seems that particle diversity is generally more equitable in *North* and *North-East* regions, which could reflect the influence of the productivity; *South-West* stations likely have one or two types of particle types that predominate.

These observations might reflect the influence of the branch of the Labrador current flowing from the North through the margin trough and interacting with the bathymetry, which could bring and maintain high densities of particles and plankton.

Recommendations

1. It is recommended to take note from all possible resources about UVP, Ecopart and Ecotaxa functioning. Ideally, users should receive training about the use of UVP and Ecotaxa from the team of Villefranche-sur-Mer.
2. It is necessary to perform image prediction on Ecotaxa by always using the same learning set, adapted to the region and with the same categories. Details are given in the guide associated with this report. Otherwise, profiles and areas will not be comparable with each other over time.
3. Score 1 (quantity of particles and plankton) is quite reliable as it is based on 3 robust metrics: however, it can be highly spatiotemporally variable according to season and patchiness.
4. Score 2 (diversity of particles and plankton) can be difficult to interpret and it is recommended to look for the influence of the different particle types at each station (see supplementary plots produced in the code).
5. For a better characterization of pelagic biodiversity, manual validation of images into categories would be highly beneficial. After manual validation, some other biodiversity indices with higher ecological significance could be computed, such as species richness, Shannon diversity index of species, or the distribution of rarer plankton species than copepods (i.e. cnidarians, ctenophores, annelids, appendicularians).

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